

3a: Direct Orthogonal Feature Correction Protocol

What to investigate, what to learn, and how to run the first study

Research working guide | 8 September 2026

Primary question. Can a single orthogonal transform learned from matched real 30 m and real 80 m calibration pairs map held-out real 80 m features into the 30 m feature space well enough to improve an unchanged linear probe trained only at 30 m?

1 What this experiment is meant to establish

Question 3a is a representation experiment. It asks whether altitude produces a sufficiently consistent displacement in a frozen vision model’s feature space that one shared rotation or reflection can correct it. The result is judged on *new physical objects*, not on the pairs used to estimate the matrix.

The experiment does not initially require synthetic imagery, detector retraining, or an onboard flight. Those are later practical or mechanism studies. It first needs a clean comparison between the same fixed 30 m probe on: (i) uncorrected real 80 m features and (ii) corrected real 80 m features.

Three terms must stay precise:

- **Frozen encoder:** the vision foundation model and the selected feature layer do not change.
- **Fixed 30 m probe:** the classifier is fitted using 30 m training objects only; its weights, intercept and decision rule do not change during 80 m evaluation.
- **Target-calibrated correction:** the Procrustes matrix uses some matched real 80 m calibration data. This is correction without retraining the encoder or probe, but it is not correction without target-altitude data.

The operational meaning of “map onto the 30 m manifold” should be defined through two tests: corrected 80 m features become closer to their paired 30 m features on held-out objects, and the fixed 30 m probe becomes more accurate on those corrected features. A low fitting error on the calibration pairs alone is not enough.

2 What to learn before implementation

Learn the topics below in dependency order. The right-hand column is the concrete output that shows enough understanding to proceed.

Topic	Minimum understanding needed	Output for this study
Camera geometry and GSD	Why object pixel size changes with height; assumptions behind the 30/80 scale ratio; effects of terrain, oblique view and camera settings	A camera sheet and measured object-size ratios from paired images

Cross-altitude matching	Homography for near-planar nadir scenes, its failure on multiple depth planes, and how to verify that two patches contain the same physical object	A montage of accepted and rejected 30/80 pairs with registration error
Frozen vision features	Encoder input preprocessing, layer choice, CLS versus patch tokens, pooling and feature normalization	A written feature specification that produces one fixed-length vector per patch
Linear probing	Multiclass logistic regression or a linear classifier, regularization, class imbalance and validation without changing the encoder	A reproducible 30 m probe and a 30 m held-out reference score
Orthogonal Procrustes	Centering, singular value decomposition, orthogonal matrices, mapping direction and what distance preservation implies	A tested function that estimates 80-to-30 mapping and reproduces a toy example
Rank and sample size	A centred matrix with n pairs has rank at most $n - 1$; a feature dimension much larger than the number of calibration objects makes the full rotation weakly identified	A singular-value plot and a preselected reduced feature dimension
Grouped evaluation	Why video frames and multiple crops are not independent; how object, location and session leakage inflate results	A locked split manifest created before feature extraction
Paired uncertainty	Why corrected and uncorrected predictions must be compared on the same test objects; how to report effect size and uncertainty	A paired difference with a confidence interval resampled by physical object

Useful starting references are the original Schönemann Procrustes paper, the official SciPy definition, the DINOv3 paper for frozen global and dense feature probes, the OpenCV homography explanation, and the official GroupKFold documentation.

3 Decisions to fix before collecting the main dataset

3.1 Choose the first task

Use **oracle-region object classification** for the first Question 3a experiment. A human annotation supplies the object centre or box; a fixed patch is extracted; the frozen encoder produces a feature vector; and a linear probe predicts the class. This isolates feature correction from object proposal and localization errors.

This is not yet a complete detector. A later detector experiment can establish deployment value, but adding localization now would make it unclear whether an error came from altitude-sensitive features or a failed bounding-box proposal.

3.2 Choose one encoder and one representation

Start with one accessible model rather than comparing several at once. DINOv3 is a reasonable candidate because its published evaluation explicitly uses frozen patch features for dense linear probing and a frozen global representation for image classification. Its official repository notes that pretrained weights require an access request, so DINOv2 is a practical fallback if access or compute delays the pilot.

Record the exact model, checkpoint, software commit, input size, colour conversion, normalization, feature layer, token and pooling rule. For example: “DINO family ViT, last-layer CLS token, official preprocessing, no train-time feature fine-tuning.” Do not change this definition after looking at the final test set.

3.3 Choose a patch protocol that keeps altitude visible

For the primary experiment, extract a fixed native-pixel canvas around the matched object at both altitudes. Use the same canvas dimensions for every object. The object should therefore occupy fewer pixels at 80 m. Avoid tightly cropping each bounding box and resizing it to fill the network input, because that can remove much of the scale change being studied.

Run a separate sensitivity analysis with scale-normalized crops. If Procrustes helps only in the fixed-canvas condition, the corrected component is probably tied to scale or context. If it also helps after normalization, other appearance changes may contribute.

3.4 Start with controllable objects

The first pairs should be static road markings, utility covers, signs, fixed campus objects and parked vehicles that remain in place across the two passes. Moving pedestrians and vehicles are poor pilot pairs because exact physical identity and pose cannot be held constant while the aircraft changes height.

Begin with nadir views at 30 m and 80 m using the same camera, lens, resolution and recording mode. Log altitude AGL, timestamp, exposure, ISO, shutter speed, gimbal angle, aircraft pose and object identity. Treat oblique views and the other proposed altitudes as later generalization tests.

4 The data split is part of the method

Create the split before making crops, extracting frames, generating augmentations or inspecting final results. Every image, altitude, frame and derivative of one physical object belongs to exactly one group.

Partition	Permitted use	Forbidden use
Source-train objects	Fit feature preprocessing learned from source data and train the linear probe using 30 m features and labels	Estimating Procrustes or reporting the final effect
Calibration objects	Estimate R and, for the centred secondary model, the 30 m and 80 m calibration means	Training the probe or selecting the winning method from final performance

Development objects	Select representation details, probe regularization and any reduced feature dimension; diagnose failures	Appearing in the final test or repeatedly changing choices after final evaluation
Final-test objects	Evaluate the frozen choices once on paired real 30/80 m observations	Fitting, tuning, threshold selection or error-driven redesign

Where possible, place the final-test objects in a different geographic subsection or flight session. If several patches belong to one object, calculate uncertainty by resampling whole objects, not individual patches. Non-overlapping groups are the principle implemented by grouped cross-validation tools such as GroupKFold.

The 10-object pilot in the original proposal is appropriate for checking registration, metadata, feature shapes and software flow. It is too small to decide whether Question 3 is true. The main sample target should be stated as independent objects per class and checked after the pilot for class balance and uncertainty; thousands of adjacent video frames do not replace independent objects.

5 The exact experiment

Let $f(\cdot)$ be the frozen encoder. For each object i , extract matched vectors

$$z_{30,i} = f(x_{30,i}), \quad z_{80,i} = f(x_{80,i}).$$

Fit a linear classifier g only on source-train z_{30} and labels. Freeze f and g .

5.1 Primary: strict orthogonal mapping

Stack calibration features into X_{80} and X_{30} . Estimate

$$R^* = \arg \min_{R^T R = I} \|X_{80} R - X_{30}\|_F^2.$$

If $X_{80}^T X_{30} = U \Sigma V^T$, then $R^* = UV^T$. The corrected test feature is simply

$$\hat{z}_{30} = z_{80} R^*.$$

This is the closest match to the proposal’s claim of a single orthogonal transform. Orthogonality preserves vector norms, angles and pairwise Euclidean distances under the linear map, so it cannot restore information that the 80 m image never resolved.

5.2 Secondary: centred orthogonal mapping

A common Procrustes workflow centres each domain first:

$$\hat{z}_{30} = (z_{80} - \mu_{80}) R^* + \mu_{30}.$$

Here, R^* is orthogonal, but the complete correction includes a translation. Report it separately. If this version succeeds while the strict version fails, the evidence supports a mean shift plus rotation, not a pure orthogonal correction.

The official SciPy routine minimizes $\|AR - B\|_F$ subject to $R^T R = I$ and does not add scaling or translation. Any centring, dimensional reduction or normalization must therefore be explicit in the method.

5.3 Baselines and controls

Evaluate every arm on the same final-test objects.

Arm	Operation	What it answers
A	Fixed 30 m probe on held-out real 30 m features	Source-domain reference
B	Same probe on uncorrected real 80 m features	Cross-altitude baseline
C	Strict real 80-to-30 orthogonal mapping, then the unchanged probe	Direct answer to Question 3a
D	Centred orthogonal mapping, then the unchanged probe	Whether a translation is also required
E	Mean shift only	Whether most benefit comes from domain means rather than rotation
F	Regularized unconstrained linear map using the same calibration pairs	Whether orthogonality is a useful restriction
G	Procrustes after shuffling pair identities within calibration data	Negative control for the value of exact matching

If feature dimension is large relative to calibration objects, add a reduced-space version. Fit the projection without the final test objects, choose its dimension on development objects, freeze it, and then compare all relevant arms in the same reduced space.

6 What to measure and how to decide

Choose one primary task metric before examining the final test. Balanced accuracy is straightforward when class counts differ; macro-F1 is also useful, with per-class recall reported beside it. The primary effect is the paired difference

$$\Delta = M(g(z_{80}R^*)) - M(g(z_{80})),$$

where M is the selected metric and both terms use the same held-out objects. Report a 95% confidence interval by resampling physical objects as groups. For simple per-object correctness, also report the counts corrected from wrong to right and changed from right to wrong.

Use feature diagnostics to explain the result:

- paired cosine similarity and Euclidean residual before and after mapping on calibration, development and test sets;
- singular values and effective rank of the calibration cross-covariance;
- feature norm distributions at 30 m and 80 m;
- nearest-paired-object retrieval accuracy; and
- confusion matrices and per-class changes for the fixed probe.

Call the strict Q3a result positive only if arm C improves the prespecified task metric on held-out real 80 m objects by a practically useful amount, with uncertainty reported, and the held-out alignment diagnostics move in the expected direction. Calibration residual alone cannot establish transfer.

Interpret other outcomes directly:

- C succeeds: evidence supports one target-calibrated orthogonal feature correction for the tested encoder, layer, patch protocol, site and 30/80 m pair.
- D succeeds but C fails: a translation is needed; the strict Question 3 claim is unsupported.
- F succeeds but C and D fail: the relationship may be linear but not approximately orthogonal.
- Calibration fit improves but held-out task performance does not: the map overfits or aligns feature variation irrelevant to the class boundary.
- No arm helps: altitude may remove information, change features nonlinearly, or be confounded by matching, viewpoint or capture settings.

7 How to start the pilot

The first executable version should produce evidence that the pipeline is valid, not a claim about final accuracy.

1. Write a one-page specification containing the encoder, feature, classes, fixed patch size, strict primary map, metric and object-group split rule.
2. Select 10–20 static objects spanning at least a few classes and assign immutable object IDs before flight.
3. Capture nadir 30 m and 80 m imagery in one short session with unchanged camera settings where feasible; save complete metadata.
4. Register the views, extract fixed-canvas matched patches, and manually inspect a montage. Record rejected pairs and reasons.
5. Extract frozen features and verify shapes, determinism, norms and the expected object-size change at the actual encoder input.
6. Train the source probe on 30 m pilot training objects and confirm that the code can produce arms A and B.
7. Fit strict and centred Procrustes only on pilot calibration pairs; run the negative controls and confirm that no object crosses partitions.
8. Produce one results sheet containing split counts, feature dimensions, singular values, alignment residuals and paired probe predictions. Use this sheet to finalize the main-study protocol and sample design.

The most useful immediate learning exercise is a synthetic feature test before the flight data are ready: generate a matrix A , multiply it by a known random orthogonal matrix Q , and verify that the implementation recovers Q and lowers $\|AR - B\|_F$. Then repeat with a mean shift and confirm that the strict and centred versions behave differently. This checks the mathematics without making synthetic images part of the research evidence.

Reading notes

- P. H. Schönemann, “A Generalized Solution of the Orthogonal Procrustes Problem,” *Psychometrika*, 1966. Read for the objective, constraint and SVD solution.
- SciPy, `orthogonal_procrustes`. Read for the exact software convention and the absence of automatic scaling or translation.
- A. Siméoni et al., “DINOv3,” 2025. Read Sections 6.1, 6.2 and Appendix D for frozen patch features, CLS-token linear probing and the distinction between a linear probe and a trained detection decoder.
- OpenCV, “Basic concepts of the homography.” Read for the planar-scene assumption used in matching.
- PIX4D, “Ground sampling distance in photogrammetry.” Read for the link between flight height, camera geometry and ground distance per pixel.
- scikit-learn, `GroupKFold`. Read for non-overlapping group logic; the study’s groups are physical objects, with location or flight session added where possible.

This guide defines a study protocol. It contains no measured result and makes no accuracy or deployment claim.